Fuzzy Logic Systems

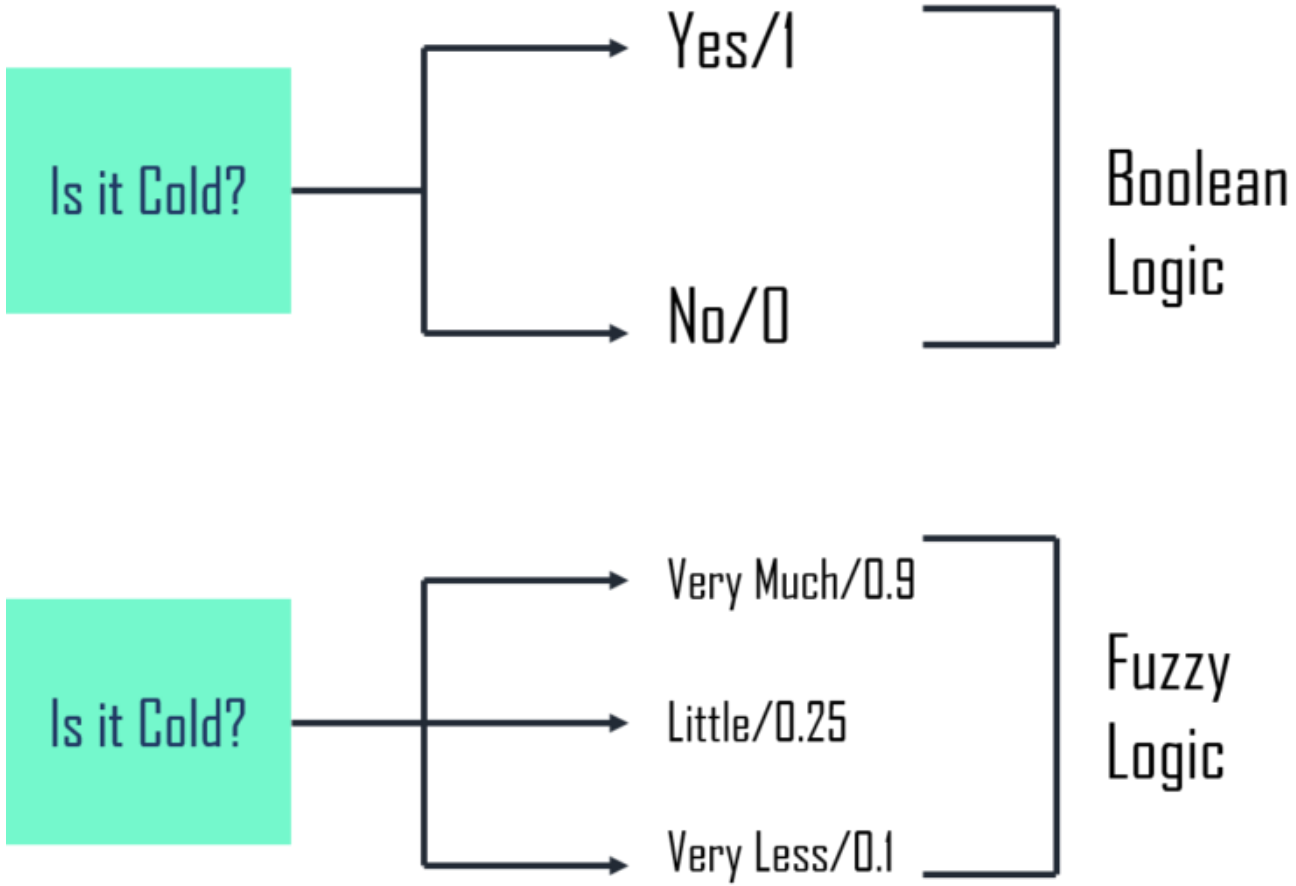
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What is it?

- A technique that allows for the inclusion of vague human assessments in computing problems
- New computing methods based on fuzzy logic can be used in the development of intelligent systems for decision making, identification, pattern recognition, optimization, and control

-Harpreet Singh, et al. 2013

Binary Vs. Fuzzy Logic

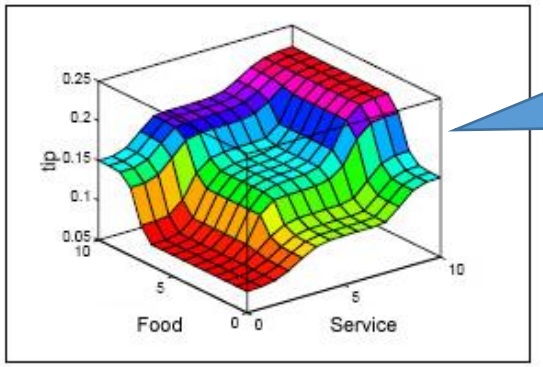


How do you give a tip at a restaurant?

- Do you measure the taste of food and quality of service, and then decide the tip?
- Or else would you rely on linguistic rules as follows?

1. *If service is poor or food is bad, then tip is cheap*
2. *If service is good, then tip is average*
3. *If service is excellent or food is delicious, then tip is generous*

- Closer to human way of thinking as well



Non-linear

Fuzzy logic

- Developed by Lotfi Zadeh in 1965
- Based on a system of non-digital (continuous & fuzzy without crisp boundaries) set theory and rules.
  - An accurate quantitative model is not required to control a plant or determine appropriate action.
  - The key advantage is its ability to deal with vague systems and its use of linguistic variables.
  - Leads to faster and simpler program development of system controllers.
- It can be a decision support system tool for managers



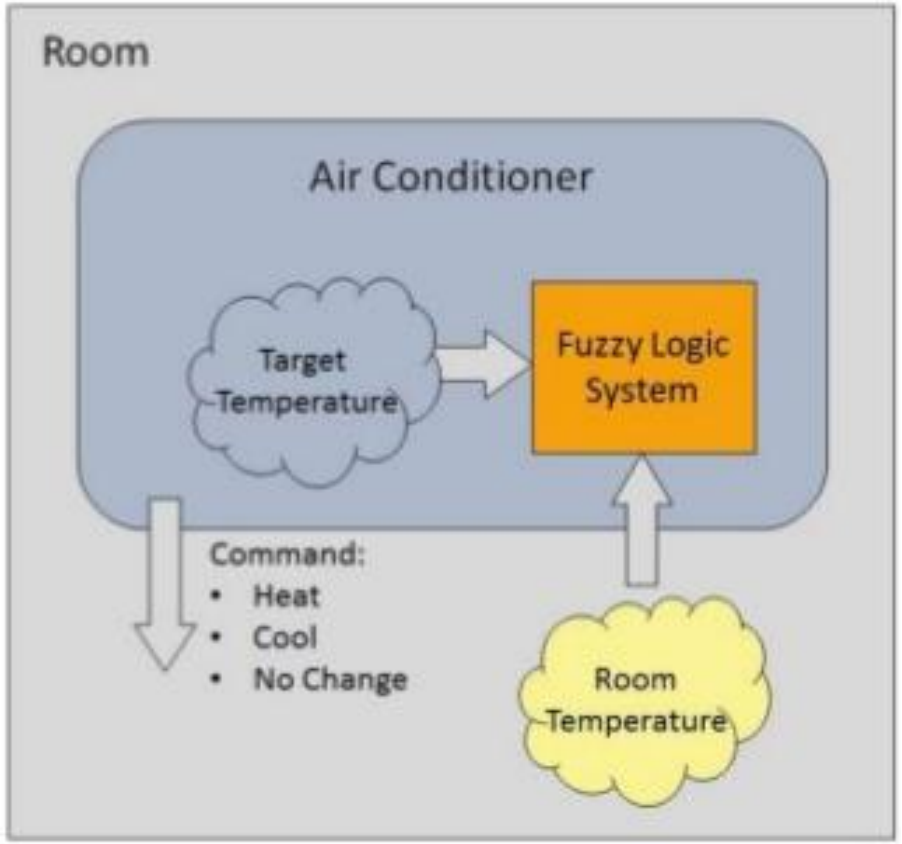
Fuzzy logic boom in Japan

- Sendai Subway automatic train operation control system – 1987
  - Fuzzy system controls acceleration, deceleration, and breaking of the train
  - Reduces energy consumption by 10%
  - Passengers hardly feel the change of velocity of the train
  - Calculations considered:
    - acceleration to a target speed with velocity oscillation
    - maintaining target speed (coasting)
    - stopping at the correct location with velocity control for a no motion feel



How fuzzy logic work in machines

- Air conditioners
  - Periodically check the room temperature against the target temperature
  - Accordingly decides whether to cool, or heat the room



Crisp sets Vs. Fuzzy sets

- Crisp set:
  - An element is either a member of the set or not
  - Most of the things that we deal with in our life belong to crisp sets
  - It’s a car, for example
  - The car is ‘Red’, is another example
- Fuzzy set:
  - An element has a partial membership to a set
  - The current distance to the car in front can be a partial member of the class ‘far’ as well as ‘close’
  - ‘Sue is tall and short’ !!

Fuzzy Logic Terminology

- Fuzzy Set: Let X be a set. X is a fuzzy set if each element y of X is characterized by a membership function $\mu_X\ y : X \mapsto [0,1]$
  - That is, $\mu X(y)$ is a value in the range 0 to 1, describing the degree of membership of y within the set X
  - Ex: A given color has 0.38 membership in the set Green
- Fuzzy Variable: a variable with fuzzy sets as its values
- Linguistic Variable: a fuzzy variable with values that are words or sentences in a language
- Linguistic Hedge: a term in a language used as a modifier of basic terms in linguistic values (e.g., very, somewhat, extremely)

Propositional Logic

- A propositional logic uses truth statements to form or provide other truth statements (applied in the forward chaining algorithm last week)
- Two types of sentences: simple and compound
  - Simple sentences are propositional constants with truth values
    - Cat is a mammal
    - All lecturers are males
  - Compound sentences are formed from simpler sentences using negations, conjunctions, disjunctions, implications, reductions and equivalences
    - Sky is cloudy AND wind is strong

Logical equivalences

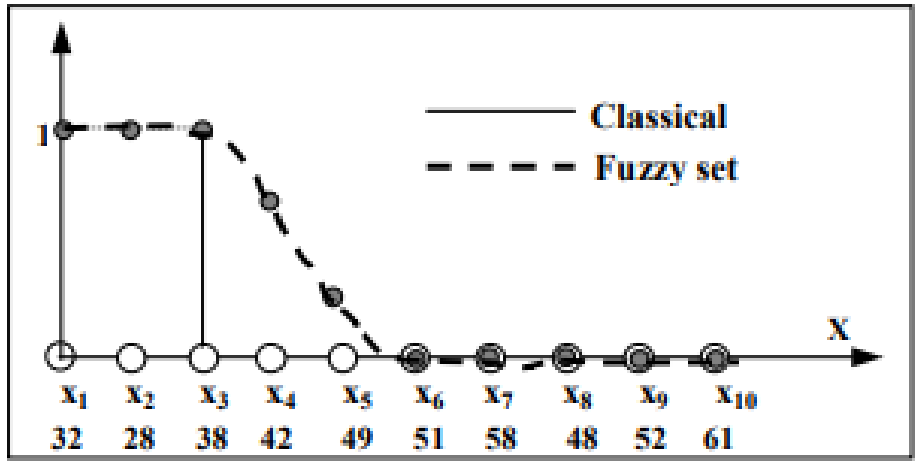
$(A \wedge B) \equiv (B \wedge A)$	$\wedge$ is commutative
$(A \vee B) \equiv (B \vee A)$	$\vee$ is commutative
$((A \wedge B) \wedge C) \equiv (A \wedge (B \wedge C))$	$\wedge$ is associative
$((A \vee B) \vee C) \equiv (A \vee (B \vee C))$	$\vee$ is associative
$\neg(\neg A) \equiv A$	Double-negation elimination
$(A \Rightarrow B) \equiv (\neg B \Rightarrow \neg A)$	Contraposition
$(A \Rightarrow B) \equiv (\neg A \vee B)$	Implication elimination
$(A \Leftrightarrow B) \equiv ((A \Rightarrow B) \wedge (B \Rightarrow A))$	Biconditional elimination
$\neg(A \wedge B) \equiv (\neg A \vee \neg B)$	”De Morgan”
$\neg(A \vee B) \equiv (\neg A \wedge \neg B)$	”De Morgan”
$(A \wedge (B \vee C)) \equiv ((A \wedge B) \vee (A \wedge C))$	Distributivity of $\wedge$ over $\vee$
$(A \vee (B \wedge C)) \equiv ((A \vee B) \wedge (A \vee C))$	Distributivity of $\vee$ over $\wedge$

Fuzzy Propositional Logic

- The membership degree $\mu_X\ y$ represents the truth value of the sentence y in the universe of discourse X
  - 0 represents absolute falsity
  - 1 represents absolute truth
- A fuzzy proposition is a sentence such as “Joe is short” with a fuzzy truth value $\mu_{short}(Joe)$
- We can also read this as: Joe is a member of the set of short people with degree $\mu_{short}(Joe)$

Classical sets Vs. Fuzzy sets

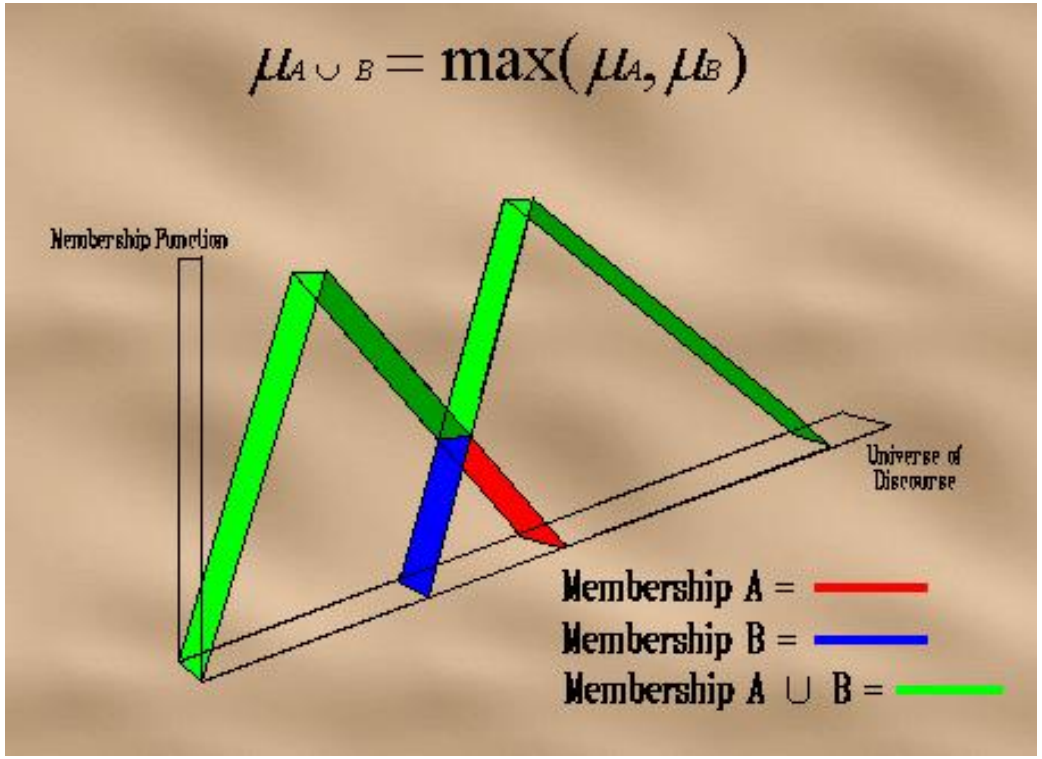
- Set A: Faculty members of a department
- Set B: Faculty members who are young (i.e., below 40)
- In traditional crisp sets:
  - Set B only contains {x1, x2, x3}
- In fuzzy implementation:
  - X4 and x5 could be defined as having partial memberships



$$\mu_B(x) = \frac{1}{x_1} + \frac{1}{x_2} + \frac{1}{x_3} + \frac{0.7}{x_4} + \frac{0.3}{x_5}$$

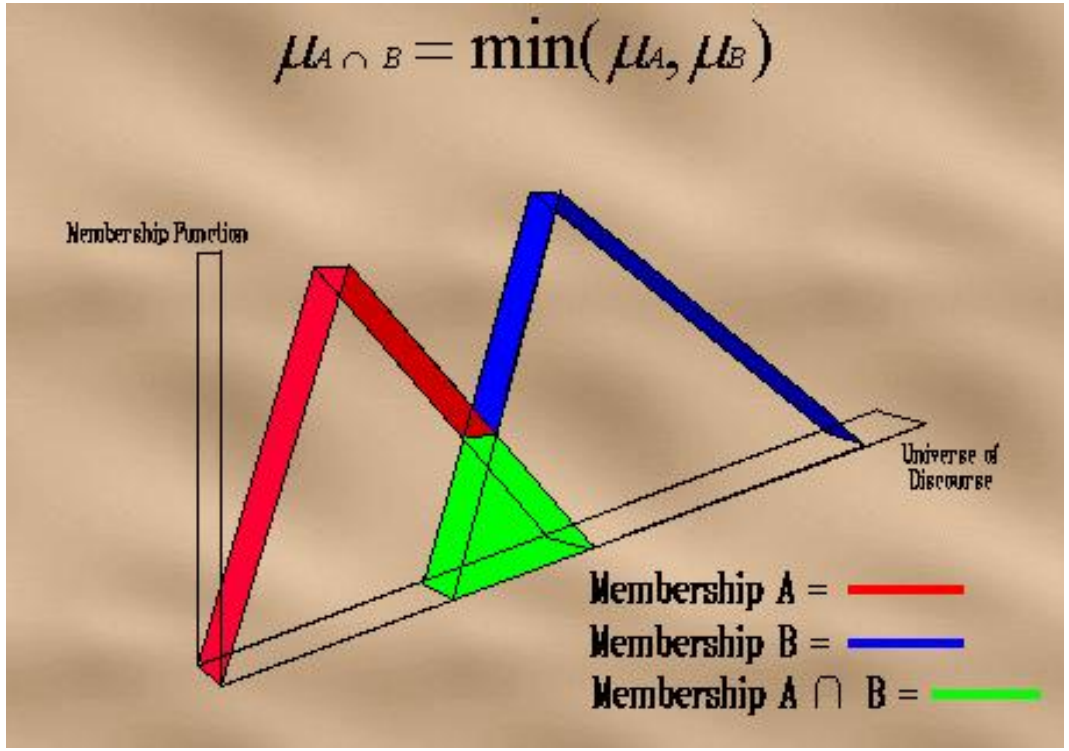
Fuzzy set operations

- Union



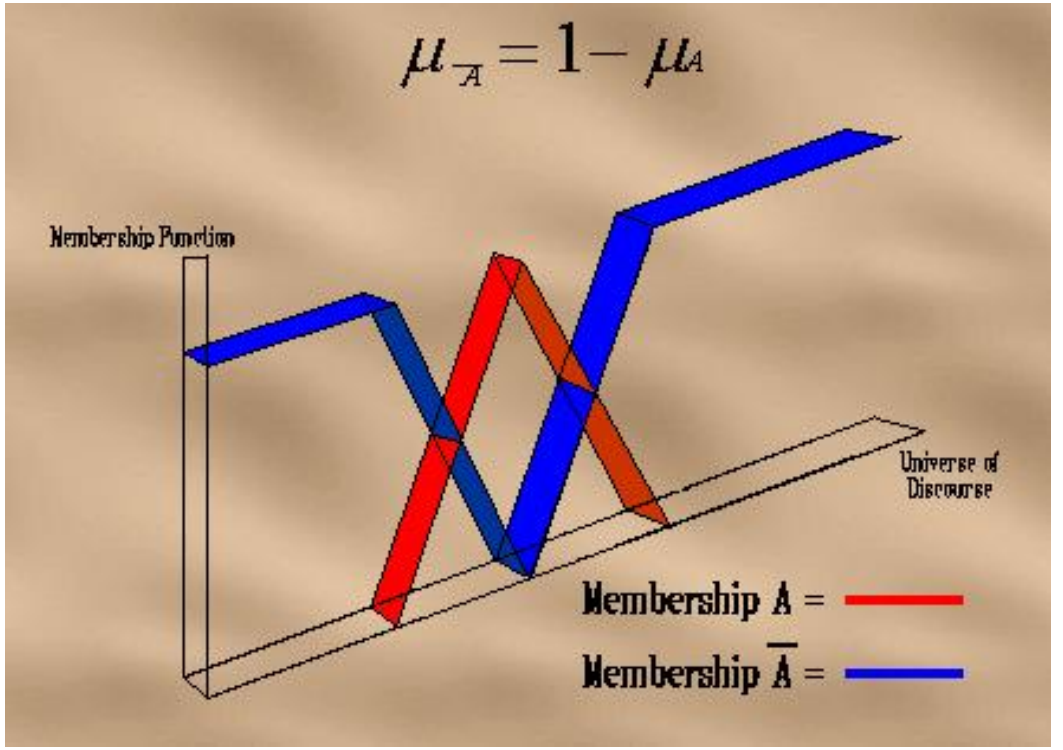
Fuzzy set operations

- Intersection



Fuzzy set operations

- Complement

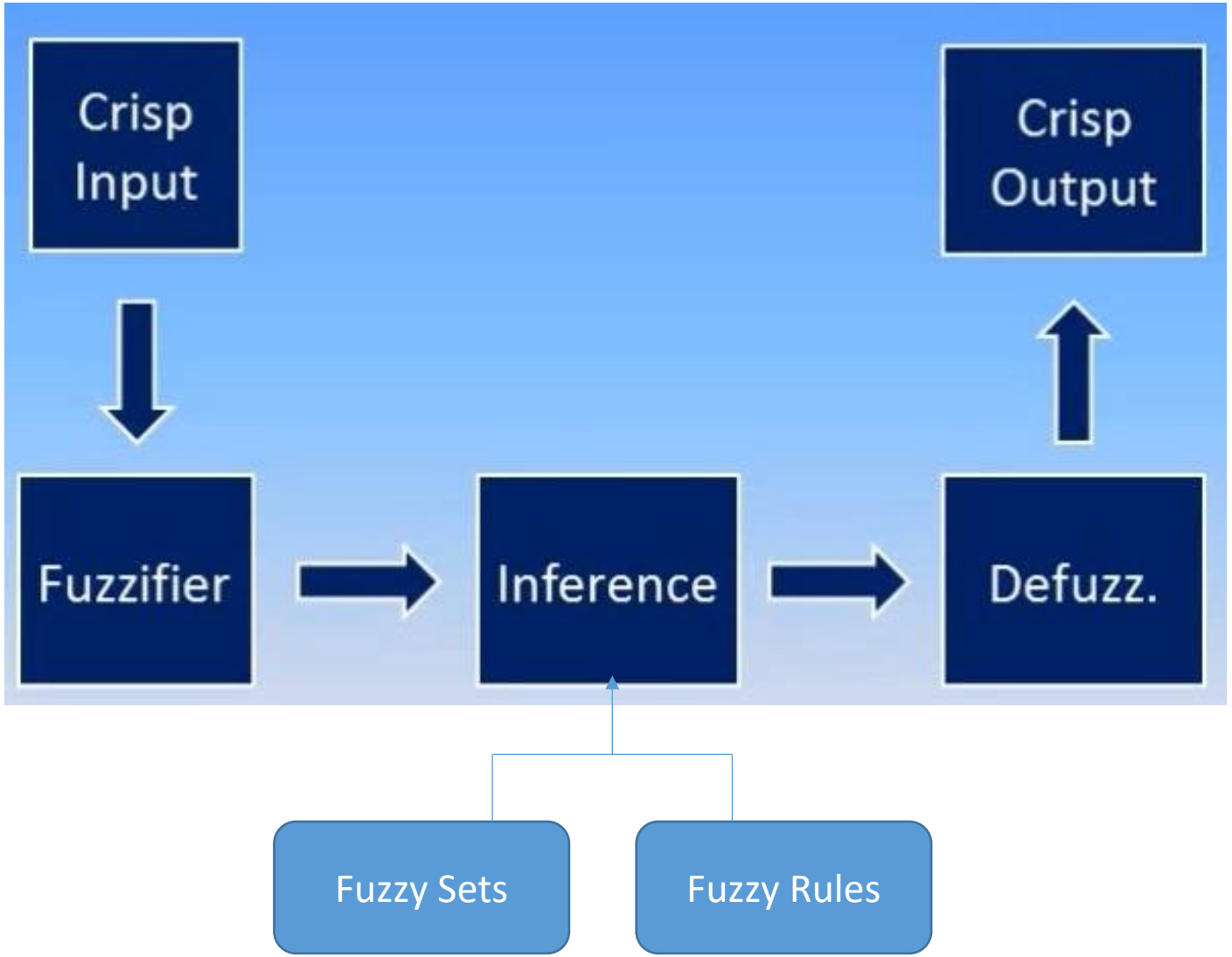


Fuzzy Rules

If antecedent(s) Then consequent(s)

- The antecedents of a rule form a combination of fuzzy sets through application of the logic operators (i.e. complement, intersection, union)
  - Requires a/the crisp input(s) to be fuzzified first
- The consequent part of a rule is usually a single fuzzy set, with a corresponding membership function

Fuzzy system components



Decision process of the air conditioner

Prepared by human experts

Table 1: Sample fuzzy rules for air conditioner system

	Fuzzy Rules
1.	IF (temperature is *cold* OR *too-cold*) AND (target is *warm*) THEN command is *heat*
2.	IF (temperature is *hot* OR *too-hot*) AND (target is *warm*) THEN command is *cool*
3.	IF (temperature is *warm*) AND (target is *warm*) THEN command is *no-change*

Fuzzy associative matrix

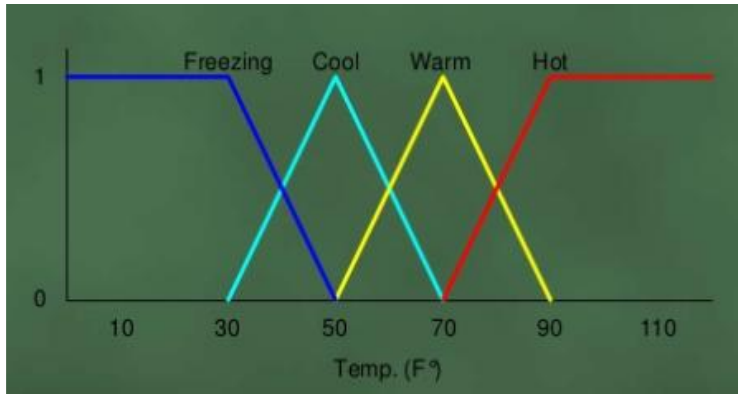
Table 2: Fuzzy matrix example

temperature/target	too-cold	cold	warm	hot	too-hot
too-cold	no-change	heat	heat	heat	heat
cold	cool	no-change	heat	heat	heat
warm	cool	cool	no-change	heat	heat
hot	cool	cool	cool	no-change	heat
too-hot	cool	cool	cool	cool	no-change

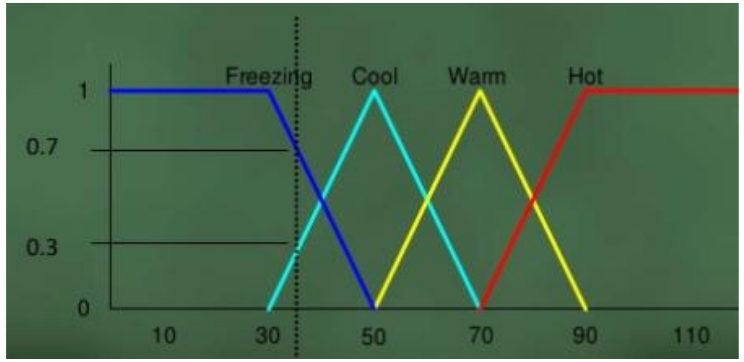
Can go up to as many as N variables

Fuzzification

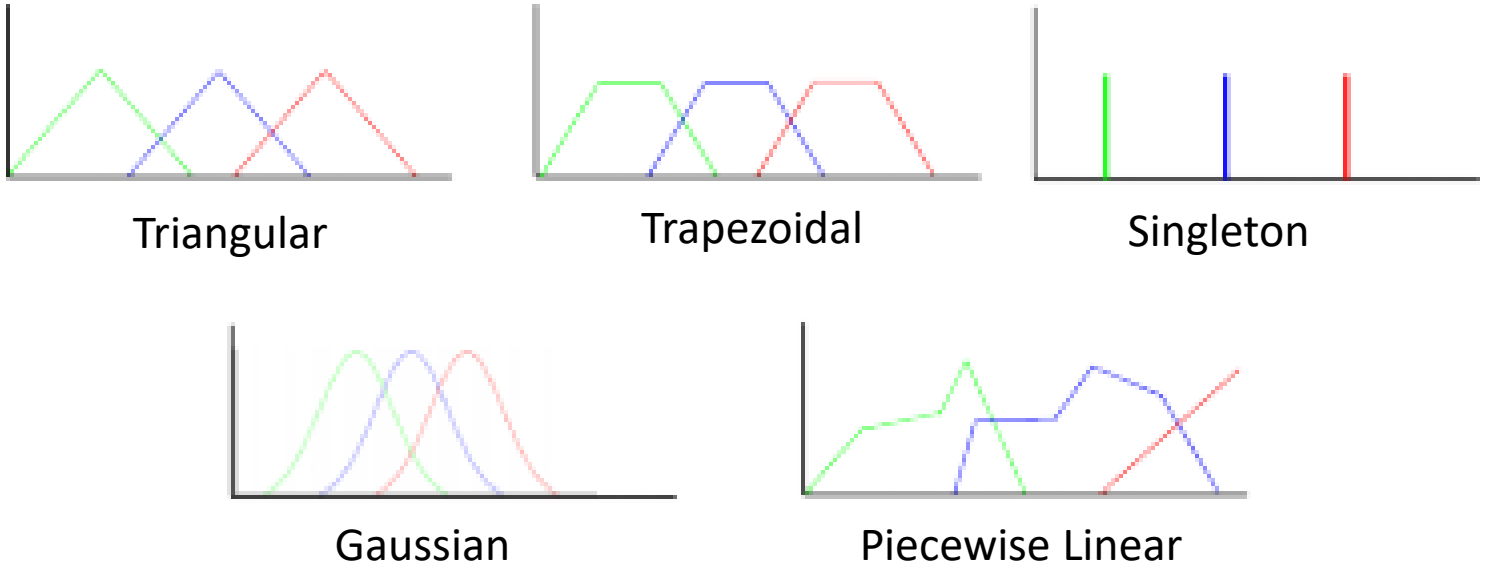
- Converting the crisp input values into fuzzy sets (linguistic variables)
  - Done through membership functions
  - Ex. Temperature is mentioned as Freezing, cold, warm or hot
  - Belongingness to a particular set cannot be defined by a threshold point
  - Fuzzification involves defining how much a particular value belongs to a given linguistic variable



How cool is 36 F°



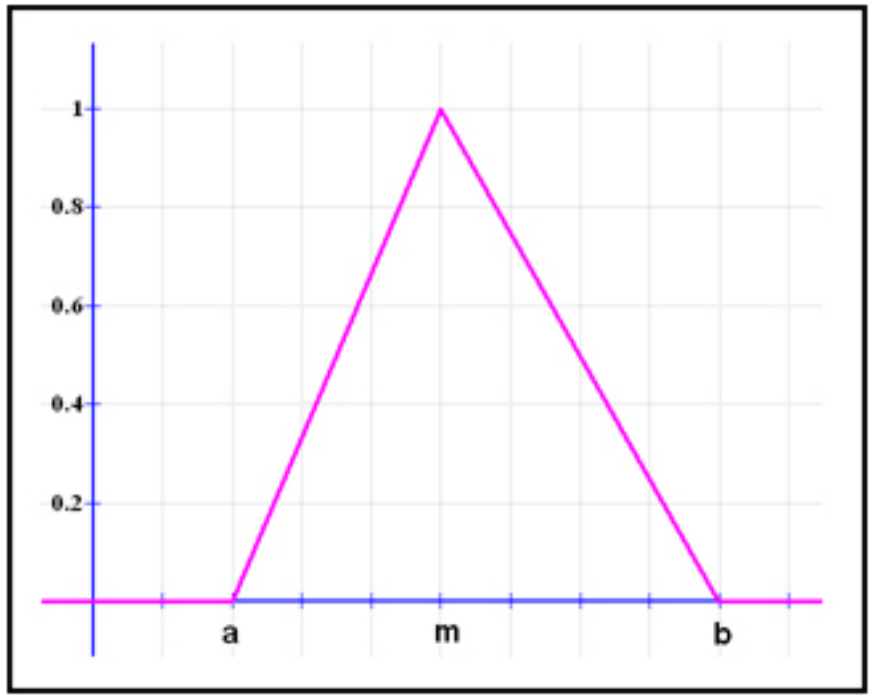
Different types of membership functions



Triangular Membership Function

Triangular function: defined by a lower limit **a**, an upper limit **b**, and a value **m**, where **a < m < b**.

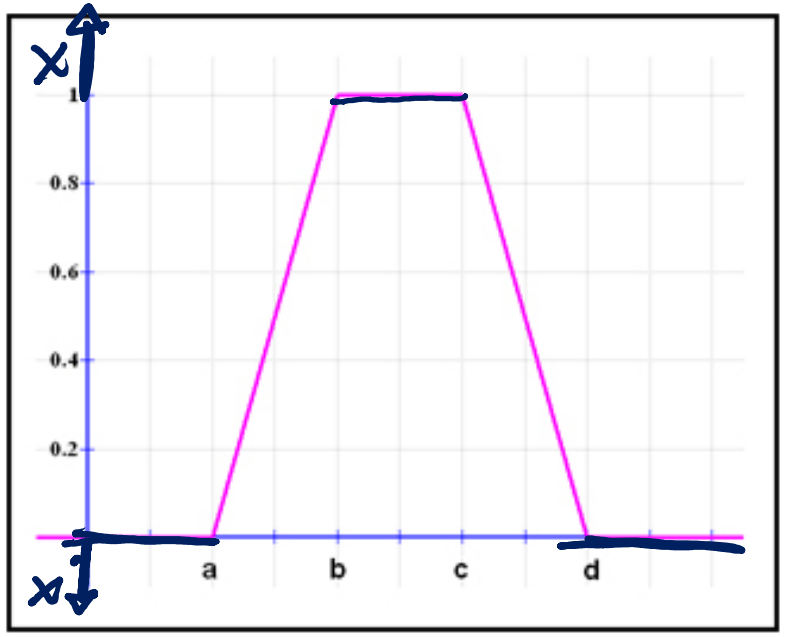
$$\mu_A(x) = \begin{cases} 0, & x \le a \\ \dfrac{x-a}{m-a}, & a < x \le m \\ \dfrac{b-x}{b-m}, & m < x < b \\ 0, & x \ge b \end{cases}$$



Trapezoidal Membership Function

Trapezoidal function: defined by a lower limit **a**, an upper limit **d**, a lower support limit **b**, and an upper support limit **c**, where **a < b < c < d**.

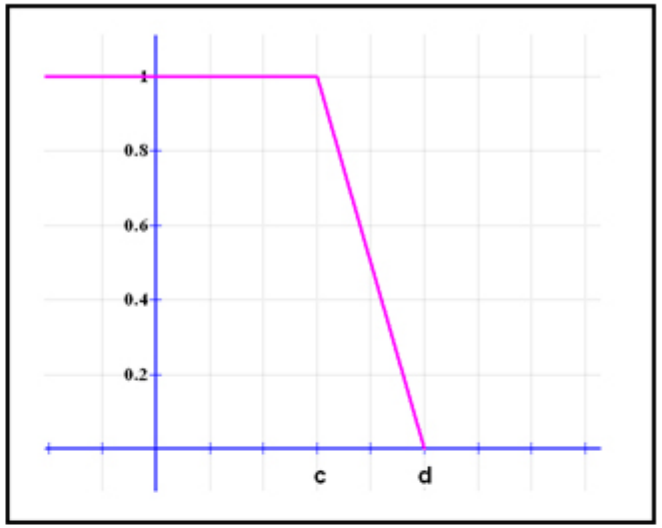
$$\mu_A(x) = \begin{cases} 0, & (x < a) \text{ or } (x > d) \\ \dfrac{x-a}{b-a}, & a \le x \le b \\ 1, & b \le x \le c \\ \dfrac{d-x}{d-c}, & c \le x \le d \end{cases}$$



L- and R- Functions

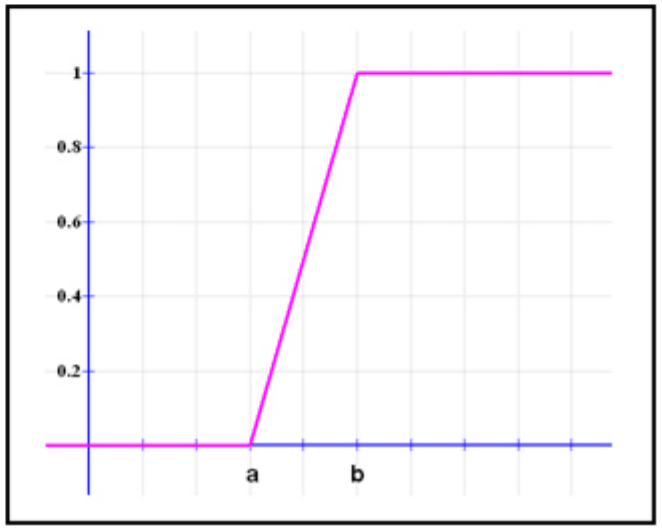
- R-functions: with parameters a = b = - ∞

$$\mu_A(x) = \begin{cases} 0, & x > d \\ \dfrac{d-x}{d-c}, & c \le x \le d \\ 1, & x < c \end{cases}$$



- L-Functions: with parameters c = d = + ∞

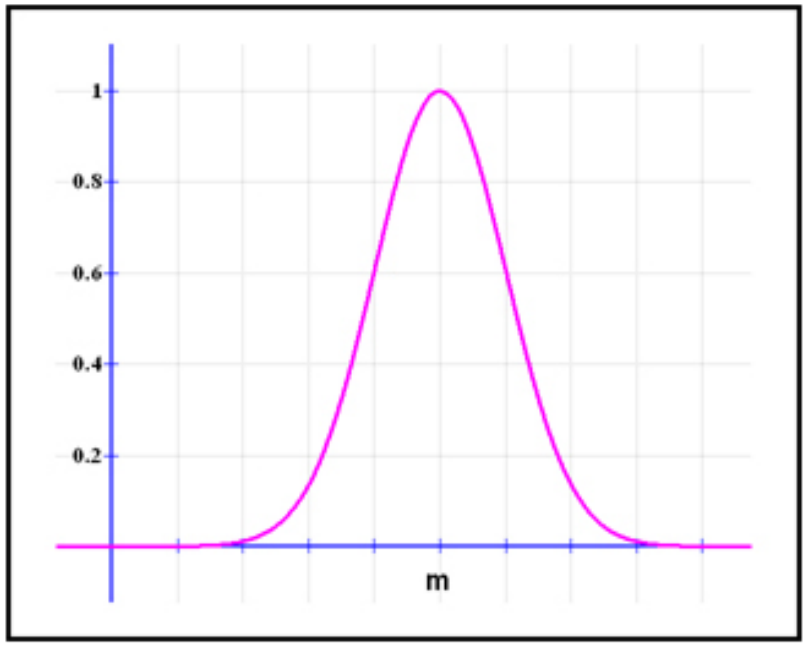
$$\mu_A(x) = \begin{cases} 0, & x < a \\ \dfrac{x-a}{b-a}, & a \le x \le b \\ 1, & x > b \end{cases}$$



Gaussian Function

Gaussian function: defined by a central value **m** and a standard deviation **k > 0**. The smaller k is, the narrower the "bell" is.

$$\mu_A(x) = e^{-\frac{(x-m)^2}{2k^2}}$$



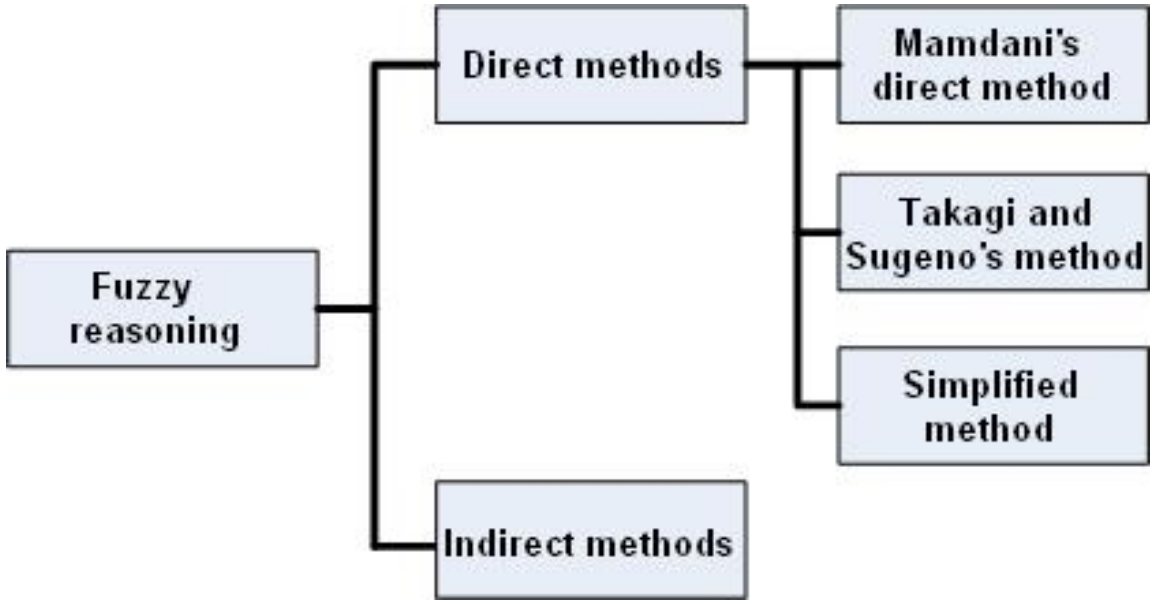
Inference

- Set of rules to be evaluated to derive the output
- Evaluation of rules require fuzzy set operations

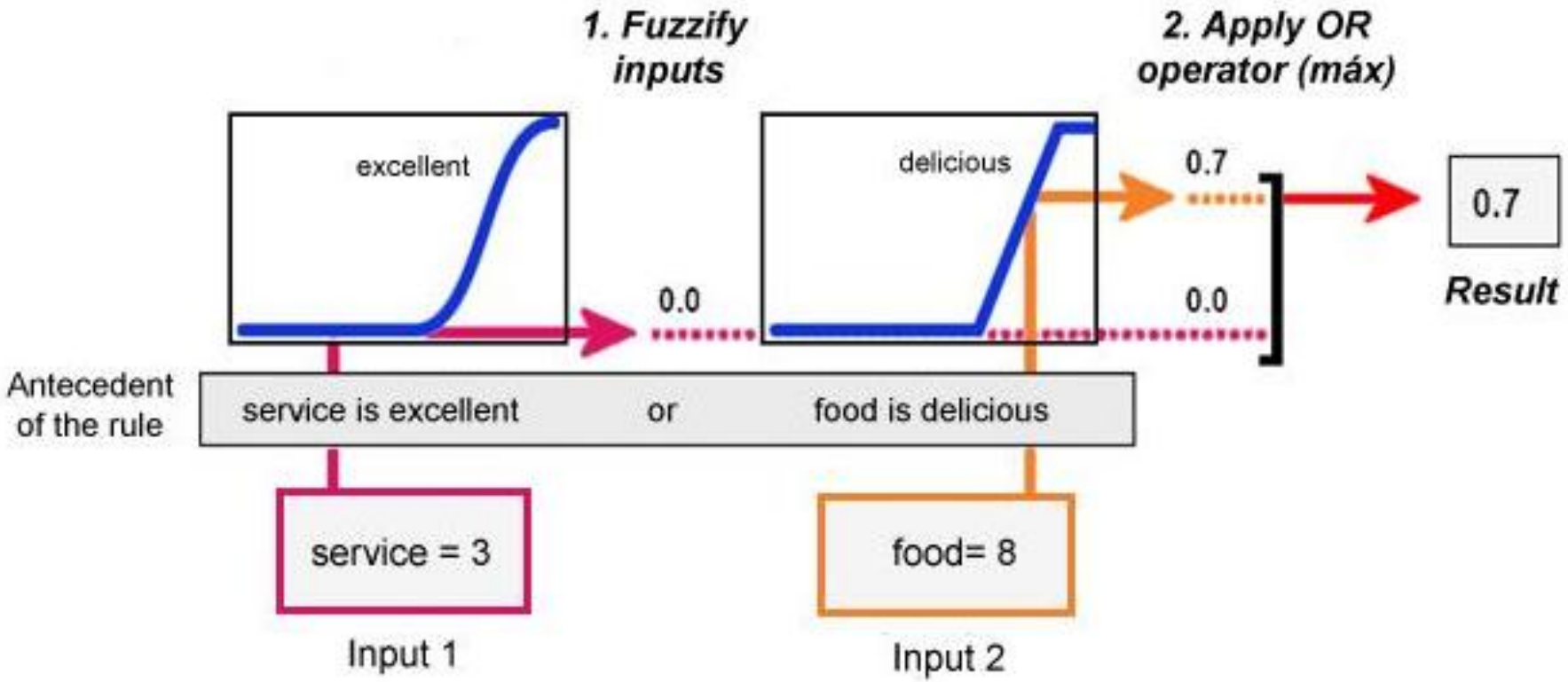
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3.	IF (temperature is *warm*) AND (target is *warm*) THEN command is *no-change*

Fuzzy inference methods

- Most commonly used methods are the Madmani's method and the Sugeno's method.
  - Mainly differ in the way they take the output



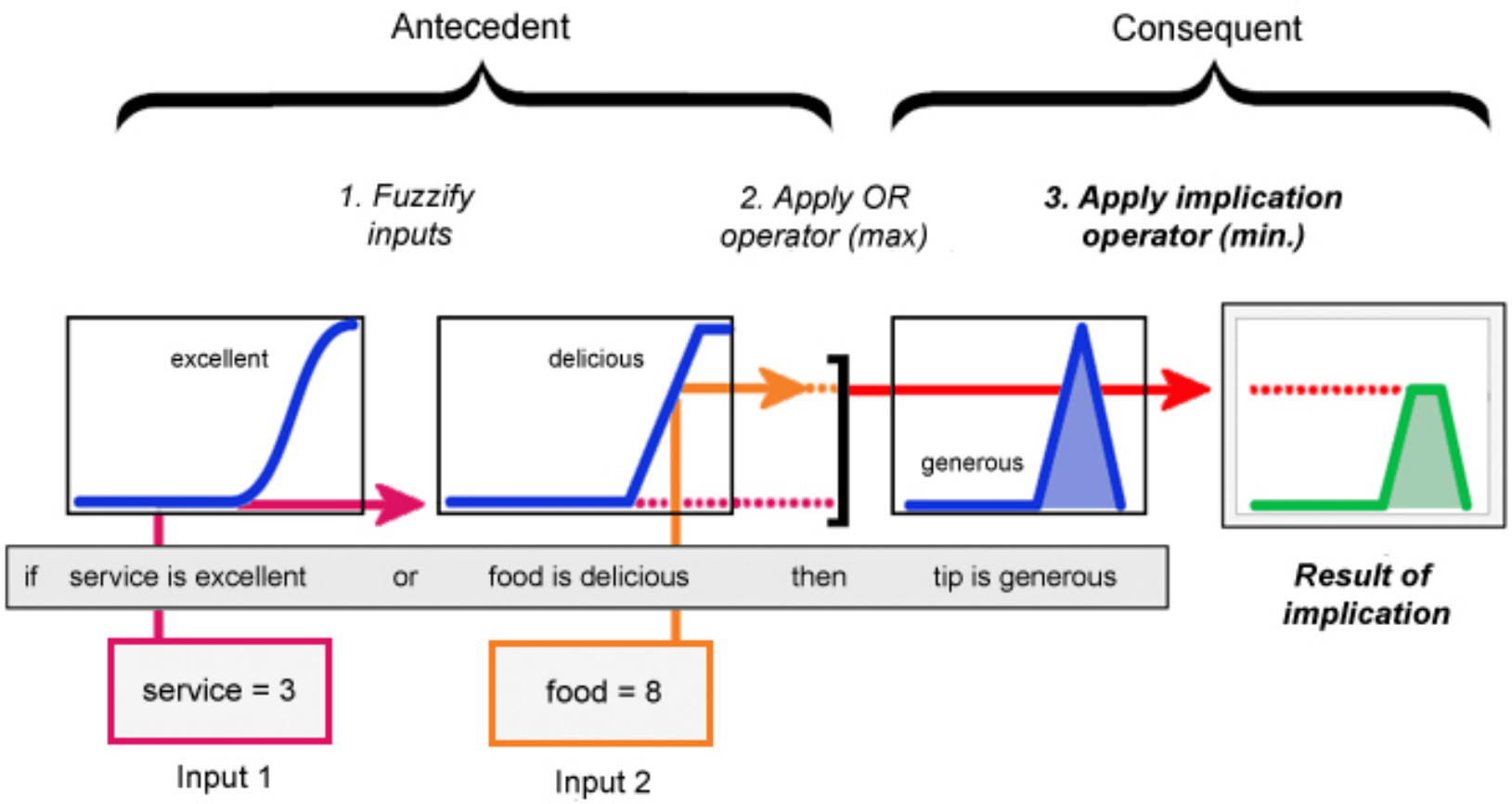
Evaluating the antecedent – Mamdani's method



Applied for each rule

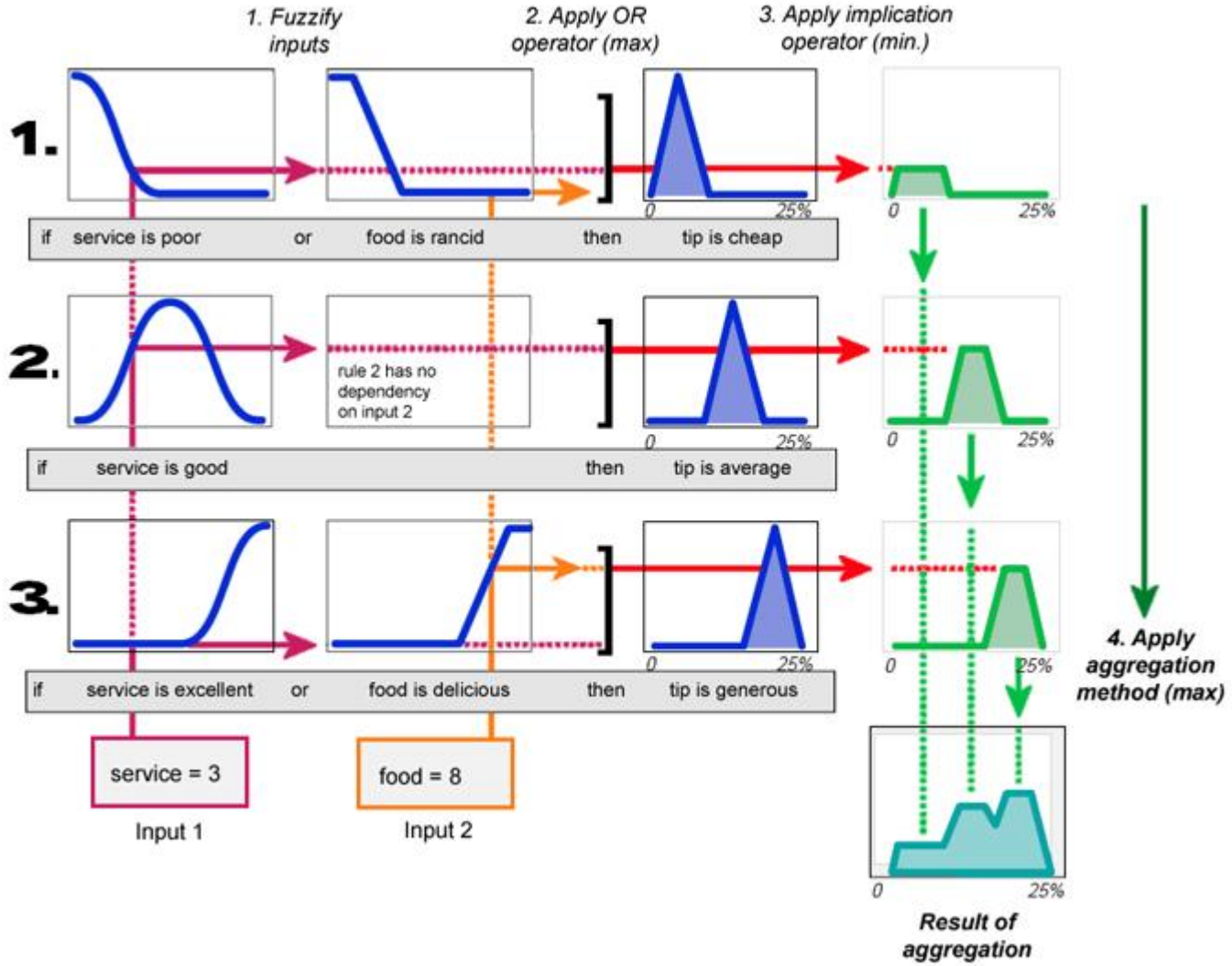
Obtaining the conclusion

- Uses the minimum which truncates the membership function.
  - Can also use product, which multiplies the membership function

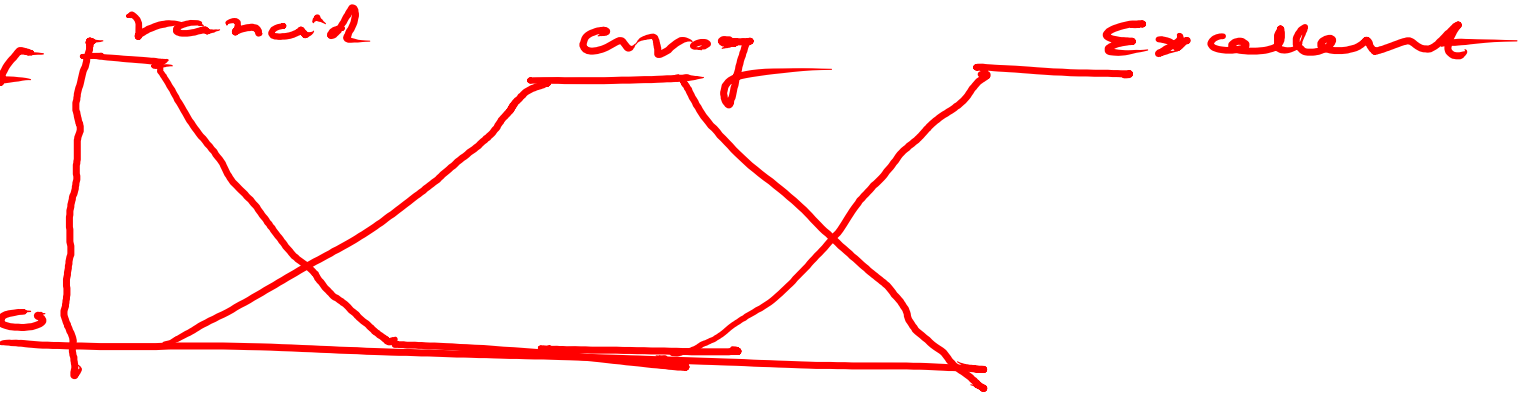


Combining multiple rules

- Some of the most commonly used aggregation operators are the maximum, the sum and the probabilistic sum
- Ex: Using max

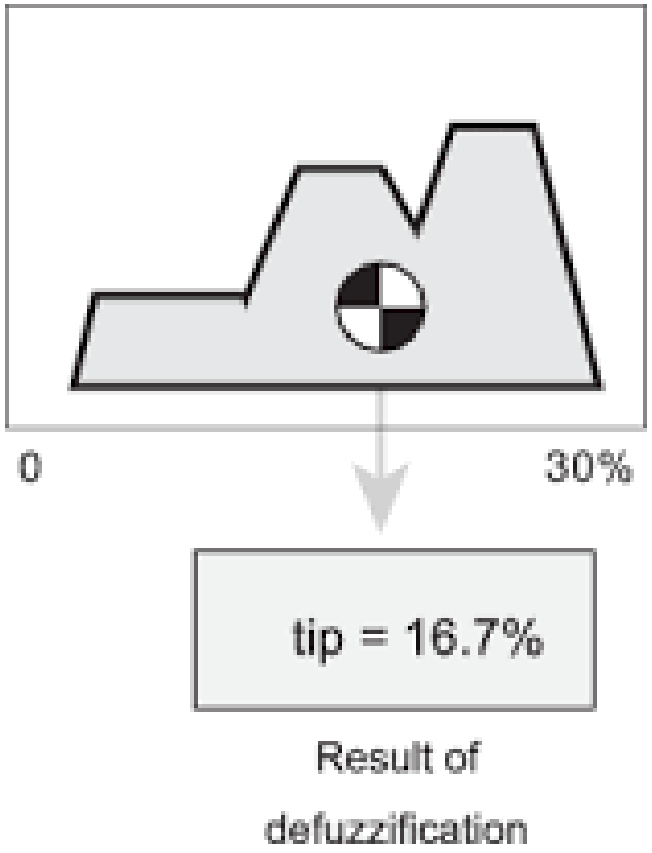



poor
Good
Excellent


rancid
Excellent

Defuzzification

- The result of inference (control output) is still vague or fuzzy
- Defuzzification helps getting an output suitable for real applications from a fuzzy inference
- Convert the fuzzy output back to the crisp or classical output to the control objective
- There are few notable methods:
  - first of maximum, middle of maximum, last of maximum, mean of maximum, center of gravity (centroid)



5. Defuzzify the aggregate output (centroid).

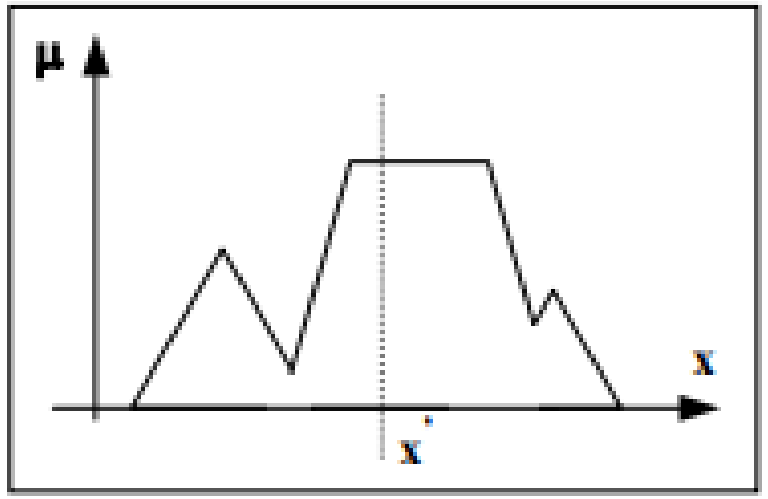
Defuzzification Techniques

- Center of Gravity Method (Centroid method)
  - Could be considered as the most popular technique
  - The weighted average of the membership function or the area bounded by the membership function curve is computed

$$\text{COG(FAST)} = \frac{\sum_x \mu_{FAST}(x) \times x}{\sum_x \mu_{FAST}(x)}$$

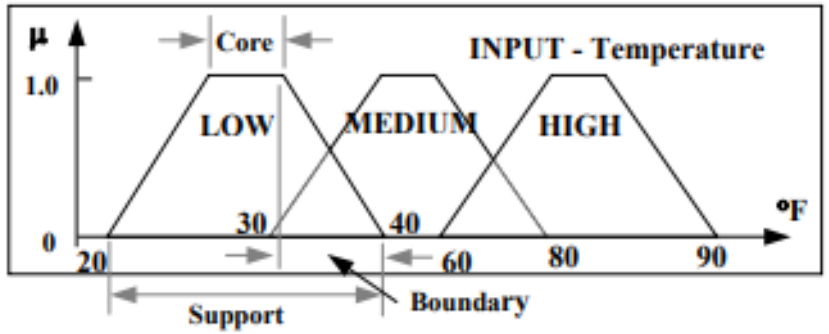
  - If x is continuous

$$\text{COG(FAST)} = \frac{\int \mu_{FAST}(x) x dx}{\int \mu_{FAST}(x) dx}$$

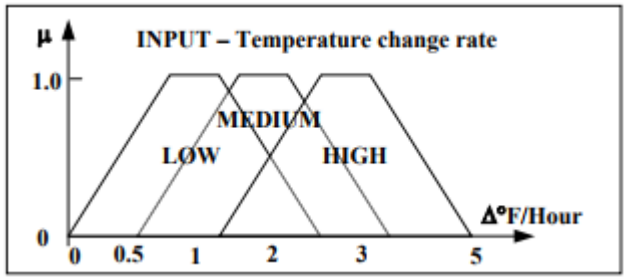


Example

- Suppose an air conditioner operates with two inputs; the current temperature and the temperature change rate (change per hour)
- There are two input membership functions



Membership function of temperature input



Membership function of temperature change input

Example

- Fuzzy control rules could be represented as a table

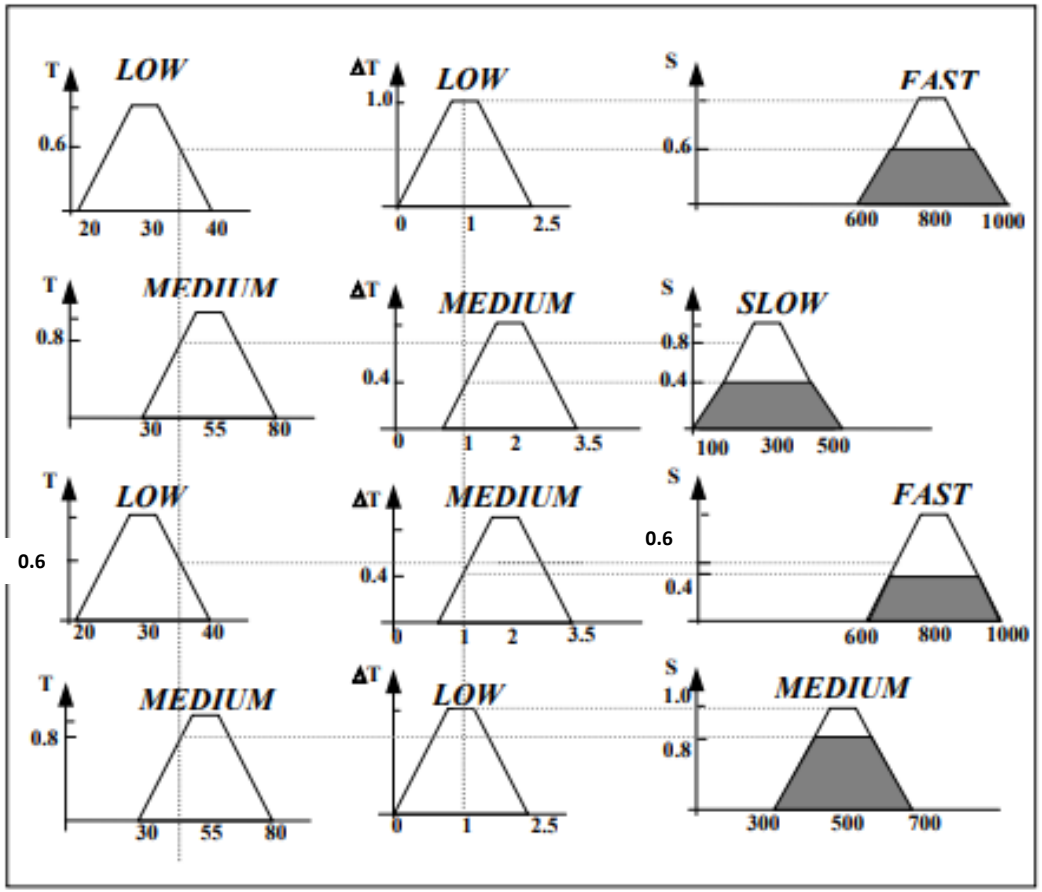
$\dot{T}$ \ T	LOW	MEDIUM	HIGH
LOW	FAST	MEDIUM	MEDIUM
MEDIUM	FAST	SLOW	SLOW
HIGH	MEDIUM	SLOW	SLOW

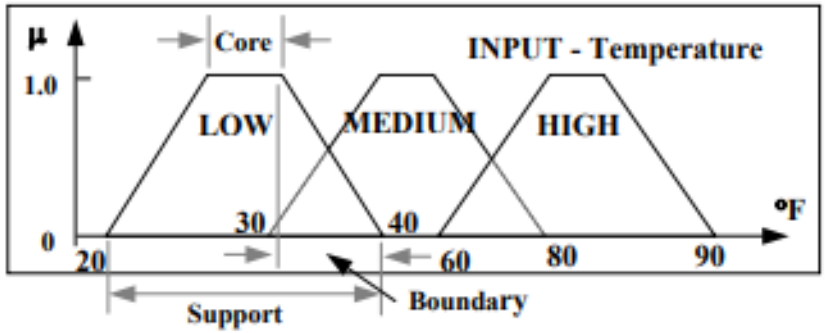
Suppose these four rules are applied

1) IF the temperature is LOW, and the change rate of the temperature is LOW, THEN the heater motor speed should be FAST
2) IF the temperature is MEDIUM, and the change rate of the temperature is MEDIUM, THEN the heater motor speed should be SLOW
3) IF the temperature is LOW, and the change rate of the temperature is MEDIUM, THEN the heater motor speed should be FAST
4) IF the temperature is MEDIUM, and the change rate of the temperature is LOW, THEN the heater motor speed should be MEDIUM

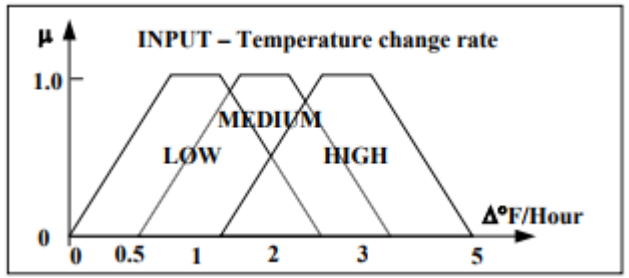
Example

- The fuzzy rules could be illustrated as functional graphs
- Consider a case of 35F^{o} temperature and and 1F^{o}/hr change of temperature
- Fuzzy output is the intersection or AND

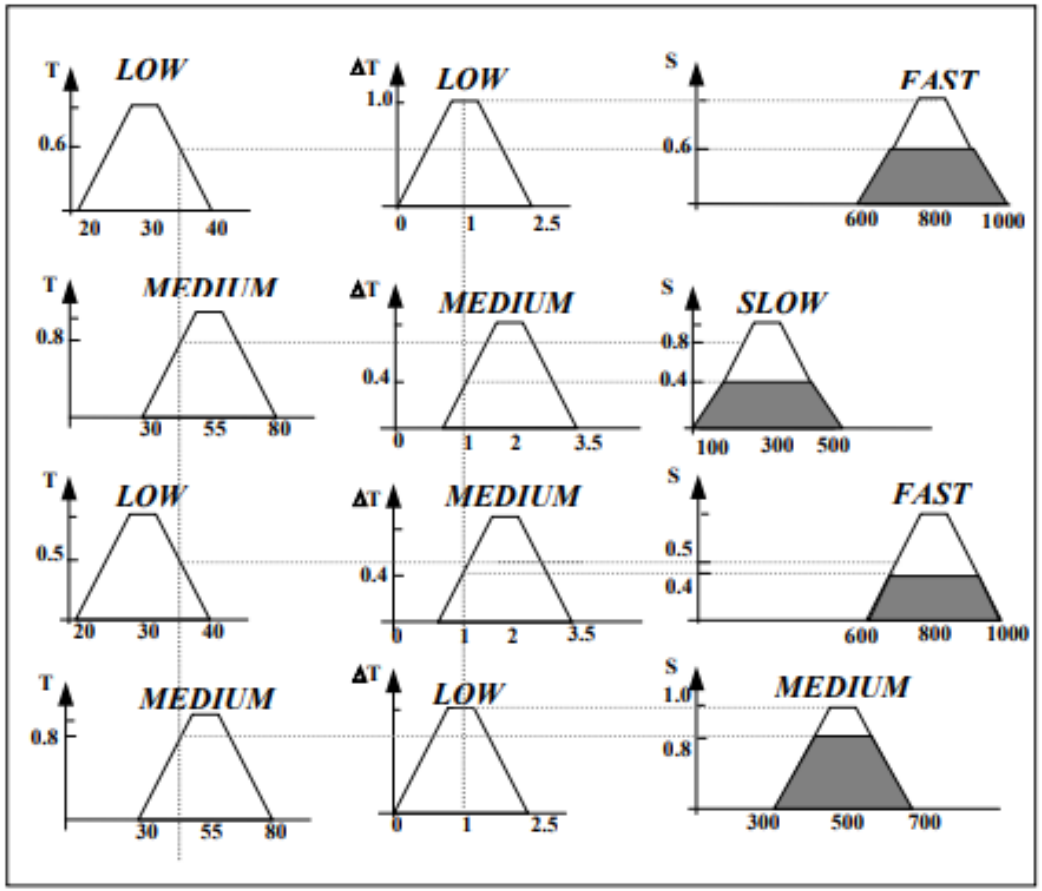




Membership function of temperature input



Membership function of temperature change input

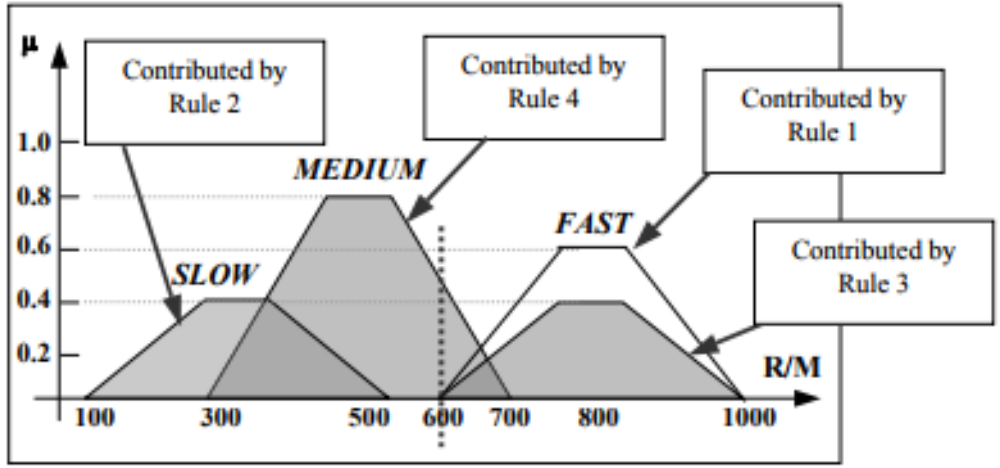


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3) IF the temperature is LOW, and the change rate of the temperature is MEDIUM, THEN the heater motor speed should be FAST
4) IF the temperature is MEDIUM, and the change rate of the temperature is LOW, THEN the heater motor speed should be MEDIUM

Input:
35F and 1F/hr

Example

- Four sets of fuzzy outputs exist for a given pair of temperature and change of temperature
- How to determine the crisp value to control the device?
- CoG method
  - Action is given by the center of the summed area, which is contributed by the different fuzzy outputs



$$y = \frac{0.6 \times 800 + 0.4 \times 300 + 0.4 \times 800 + 0.8 \times 500}{0.6 + 0.4 + 0.4 + 0.8} = 600 \text{ R/M}$$

What are the data analytics applications of Fuzzy Logic?

- Can fuzzy logic be used for image annotation?

References

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- https://jecei.sru.ac.ir/article_526_35eeea5f0f632cb283d22d78960d9308.pdf